Kafka Monitoring Alert Triage SOP

Objective

Standardize triage of Kafka alerts generated from broker, producer, consumer, and topic monitoring signals.

Scope

Applicable to Kafka-based applications, streaming platforms, backend services, data engineering pipelines, DevOps teams, and platform teams supporting production or synthetic incident workflows.

Pre-requisites

1. Access to Kafka cluster, namespace or broker host.
2. Access to monitoring dashboards such as Grafana, Prometheus, or cloud monitoring.
3. Kafka CLI tools: kafka-topics, kafka-consumer-groups, kafka-console-producer, and kafka-console-consumer.
4. Permission to view application logs, broker logs, and deployment status.
5. Escalation contact for DevOps, platform, and application owner teams.

Incident Metadata

Synthetic operational SOP derived from Kafka ticket and log patterns in the provided dataset. Use this document for alert triage, validation, and ticket enrichment scenarios.

Procedure

1. Capture alert name, severity, impacted service, topic, consumer group, and time window.
2. Correlate alert with recent ERROR, WARNING, or CRITICAL logs.
3. Map alert to known issue category such as lag, broker, disk, offset, rebalance, or replication.
4. Check dashboards for broker availability, disk, CPU, network, partition status, and consumer lag.
5. Identify whether issue is localized to one topic/group or cluster-wide.
6. Open or update the relevant ticket with log IDs and dashboard evidence.
7. Apply the matching SOP for the confirmed incident type.
8. Continue monitoring until metrics return to normal for a sustained window.

Common Commands

```
kafka-topics.sh --bootstrap-server <broker>:9092 --describe
kafka-consumer-groups.sh --bootstrap-server <broker>:9092 --describe --group <group>
kafka-broker-api-versions.sh --bootstrap-server <broker>:9092
kubectl get pods -n <namespace> | grep kafka
kubectl logs <kafka-pod> -n <namespace> --tail=100
```

Troubleshooting Matrix

Issue	Possible Cause	Resolution
Lag alert	Slow consumer	Use consumer lag SOP
Broker alert	Broker unavailable	Use broker unavailable SOP
Disk alert	Storage capacity issue	Use broker disk full SOP
Replication alert	ISR or offline partition issue	Use replication issue SOP

Best Practices

1. Correlate alerts with logs before restarting components.
2. Use consistent ticket categories and error codes.

3. Keep alert thresholds aligned with SLA impact.
4. Link SOP used in final ticket resolution.

References

Apache Kafka Docs | Internal Logs CSV | Tickets CSV | Monitoring Dashboards | Kafka CLI Runbook